

Applying Hidden-Markov-Models for Stochastic Forecasting and Optimization of a Household's Renewable Power System

Master Thesis Proposal

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Outline

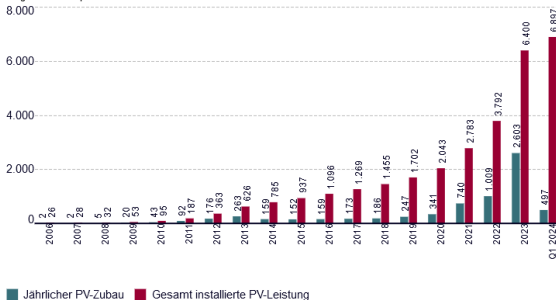
- 1 Motivation
- 2 Problem Formulation
- 3 Power Load Forecast with HMM
- 4 Validation of HMM forecast
- 5 Optimization model

Effects of energy transition for households

- Increase of power consumption, PV production, battery storages
- Causing physical problems in the distribution net
- Social acceptance inevitable

Installierte PV-Leistung in Österreich

Angaben in MWp



Quelle: BMK (2023): Innovative Energietechnologien in Österreich Marktentwicklung 2022; *vorläufige Zahlen

Household power energy system

- Components:
 - 1 grid connection
 - 2 power demand (= load)
 - 3 PV production
 - 4 battery storage
- Goal of the thesis:

Implement a stochastic load forecast and apply the prognosis to an optimization of the home energy system.

Research Question 1

What is the optimal HMM parameter setup for short-term power load forecasting?

- Properties of HMM
 - Simple, analytical structure
 - One of the first ML methods
 - Application in speech recognition
- Choice of Parameter
 - Model Size
 - Forecast horizon
 - Stochastic or deterministic forecast
- Methodology
 - Using 15min data [1]
 - Training in Julia
 - Parameter selection via Information Criterion

Research Question 2

How is the HMM prognosis performing compared to standard power load forecasting methods?

- Benchmarking models
 - Representative selection
 - Naive, kNN, MLST
- Methodology
 - Using multiple datasets
 - Validation via error metric

Research Question 3

What are the benefits of integrating stochastic load forecasts into a home energy optimization model?

- Optimization model
 - LP formulation
 - Objective: min costs
 - Decision: dis-/charging battery storage
- Methodology
 - Simulation with multiple scenarios
 - Compare different setups:
 - Active operated/passive operated/no battery storage
 - Deterministic vs stochastic forecast

Thank you!
Questions?

